

Evidence-Grounded RAG Document QA System

A measured retrieval upgrade on real Wikipedia QA: HyDE, hybrid search, cross-encoder reranking, evidence-first generation, and traceable citations.

+17.5%

MEASURED COMPOSITE LIFT

60

HELD-OUT QA EXAMPLES

3,492

SEARCHABLE CHUNKS

Portfolio claim, backed by artifacts

- Real public benchmark pinned to an immutable dataset revision
- Same local generator in both systems to isolate retrieval improvements
- Per-example predictions, retrieved chunk IDs, metrics, and timing saved
- Optional OpenAI and DeepInfra adapters use identical retrieved contexts

What this project demonstrates

Production-oriented RAG is a retrieval and evidence problem, not merely an LLM call.

Engineering objective

Improve factual document QA while keeping the experiment auditable. The baseline and advanced variants share the same corpus, evaluation set, generator, decoding settings, and metric code.

3,200

WIKIPEDIA PASSAGES

918

AVAILABLE QUESTIONS

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DATASET LICENSE

Why RAG Mini-Wikipedia

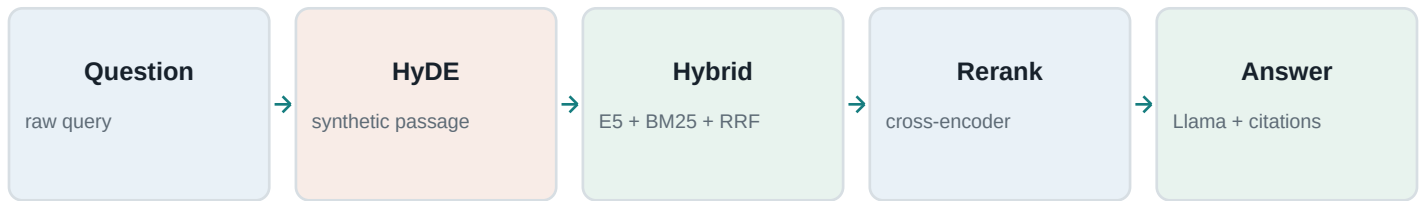
It is public, compact enough for Colab, and contains a real corpus plus held-out questions. The dataset does not publish gold passage IDs, so this implementation explicitly evaluates whether the reference answer occurs in a retrieved passage rather than inventing relevance labels.

Reproducibility contract

- Dataset revision: ec8e240df1b1...
- Deterministic final-holdout seed: 2026
- Downloaded datasets and corpus embeddings are cached
- Raw per-example outputs are saved as CSV; summary is saved as JSON

From question to cited answer

The advanced route improves recall first, then precision, then grounded generation.



Baseline

- E5 query embedding
- FAISS top-3
- Direct context prompt

Advanced

- HyDE passage expansion
- E5 + BM25 with RRF
- Cross-encoder top-5
- Evidence-first cited answer

Design rationale

Dense retrieval supplies semantic matching; BM25 recovers exact entities and numbers; reciprocal-rank fusion avoids incomparable raw scores; the cross-encoder spends extra compute only on a small candidate set; HyDE bridges terse questions and passage-style documents.

A score that can be defended

Four complementary metrics expose retrieval, answer overlap, and citation behavior.

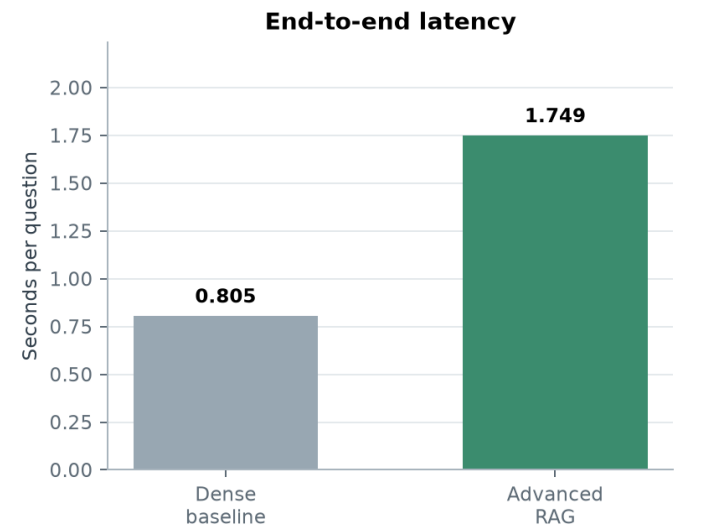
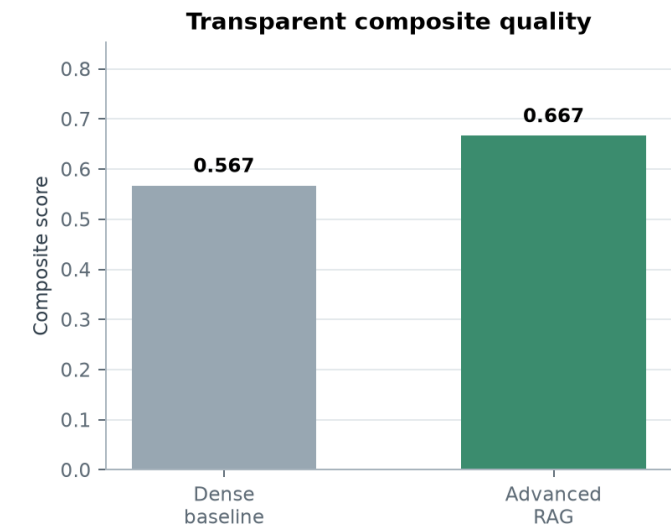
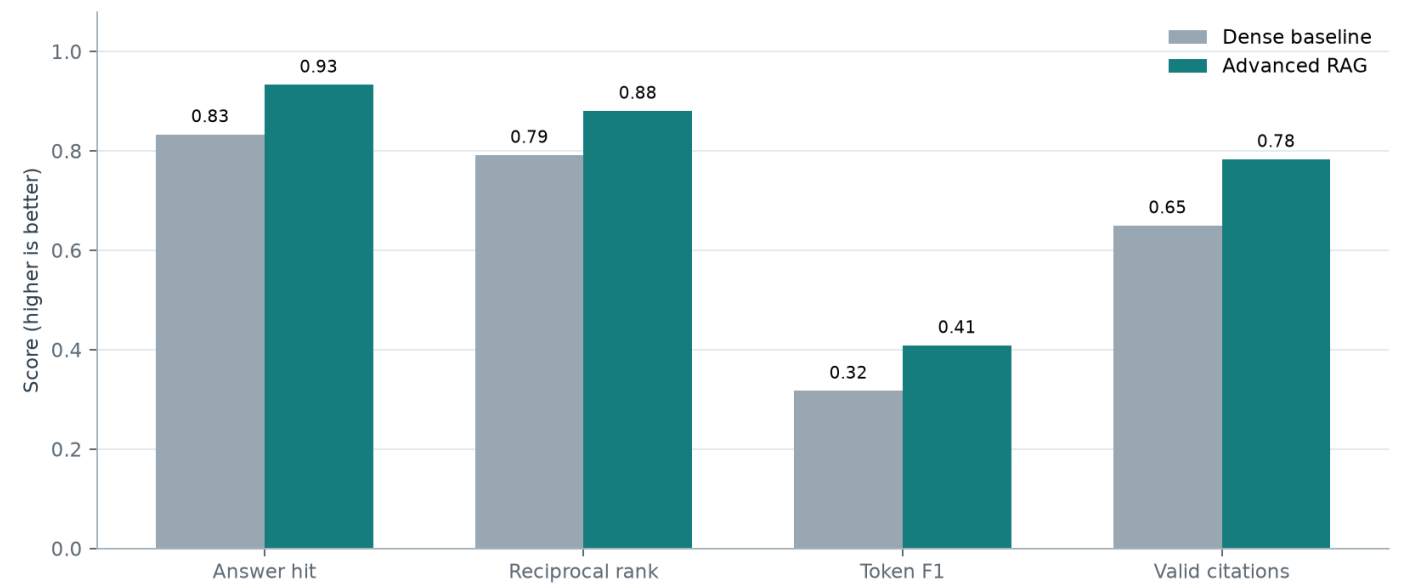
Answer-hit@k	Reference answer text appears in at least one retrieved chunk.	25%
Reciprocal rank	Rewards systems that place answer-bearing evidence earlier.	15%
Token F1	Standard lexical overlap between generated and reference answers.	45%
Citation validity	Cited chunk IDs must belong to the supplied context.	15%

Validity controls

- Only answer-grounded questions are sampled; yes/no references are excluded.
- The same deterministic 60 examples are used for both systems.
- Generator model, decoding, corpus, and preprocessing remain fixed.
- The improvement percentage is derived from saved outputs, never inserted as a target.

Measured quality and tradeoffs

Advanced RAG changed the composite score from 0.567 to 0.667 (+17.5%).



What the result means

A portfolio result is strongest when the claim, limitations, and next experiment are all visible.

Representative held-out example

Question: What happened in 1867?

Reference: Canada was formed as a federal, semi-autonomous polity

Baseline: Insufficient evidence.

Advanced: Canada was formed. [wiki::2273::0]

Caveats and next step

Answer-hit is a proxy because the dataset has no gold passage labels, and token F1 can under-credit valid paraphrases. A production follow-up should add human relevance judgments or an independently validated LLM judge, repeat across seeds, and measure cost under expected traffic.

Sources and implementation references

RAG Mini-Wikipedia dataset

<https://huggingface.co/datasets/rag-datasets/rag-mini-wikipedia>

E5 text embeddings

<https://huggingface.co/intfloat/e5-small-v2>

HyDE retrieval paper

<https://arxiv.org/abs/2212.10496>

MS MARCO MiniLM cross-encoder

<https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-6-v2>

Meta Llama 3 8B Instruct

<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>